Project: Worked examples
Title: Pipe pressure drop and pump duty
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the friction and fitting losses of series pipe segments at a given flow, the system curve and the duty point against a pump curve. It also gives the affinity-law shift for a speed or impeller change, and the NPSH.

WHAT IT CALCULATES	RELATION
Design flow and velocity	$Q = \Phi/(\rho\, c_p\, \Delta T)$ when a heat load is given in place of a flow; $v = Q/(\pi D^2/4)$
Reynolds number and friction factor	$Re = \rho v D/\mu$; $f = 64/Re$ for laminar flow ($Re < 2300$); Colebrook–White $1/\sqrt{f} = -2 \log10(\varepsilon/3.7D + 2.51/(Re\sqrt{f}))$ for turbulent flow, by fixed-point iteration (at most 100 steps) from the Swamee–Jain value, which also cross-checks it
Head lost in a segment	$h_f = f\,(L + L_{eq})/D \cdot v^2/2g + K\, v^2/2g$
System curve	$H_{sys}(Q) = \Sigma dz + h_{out} + \Sigma h_{f,lam}\cdot(Q/Q_o) + \Sigma h_{f,other}\cdot(Q/Q_o)^2$, with a laminar segment's friction scaled as Q and every other friction and fitting loss as Q^2 (a closed circuit has $\Sigma dz = 0$)
Water held in the pipes	$V_{pipe} = \Sigma\, \pi/4 \cdot D^2 \cdot L$
Duty point	$H_{pump}(Q) = H_{sys}(Q)$, every crossing found by a scan and refined by bisection, the stable one nearest the design flow reported; pump affinity laws $Q \propto n \cdot D^a$, $H \propto n^2 \cdot D^2$, with $a = 1$ for a trimmed impeller (the default) and $a = 3$ for a geometrically similar pump
Net positive suction head available	$NPSH_a = p_{atm}/\rho g + z_s - h_{f,suction} - p_v/\rho g$, with $h_{f,suction}$ that of the 'suction' segments at the larger of the design flow and the duty flow
Shaft torque for the motor	$T = P/\omega$, with P the larger of the design-flow and duty-point shaft power at the speed n_{pump}, or the shifted duty power at the shifted speed $n_{pump} \cdot n_2/n_1$

Basis: Flow of fluids through valves, fittings and pipe[1]; Internal flow: friction factor, minor losses, pump duty[2](Ch. 8, 4th ed.); Flow of fluids in pipes and ducts (principles)[3](§4); Density of liquid water[4].

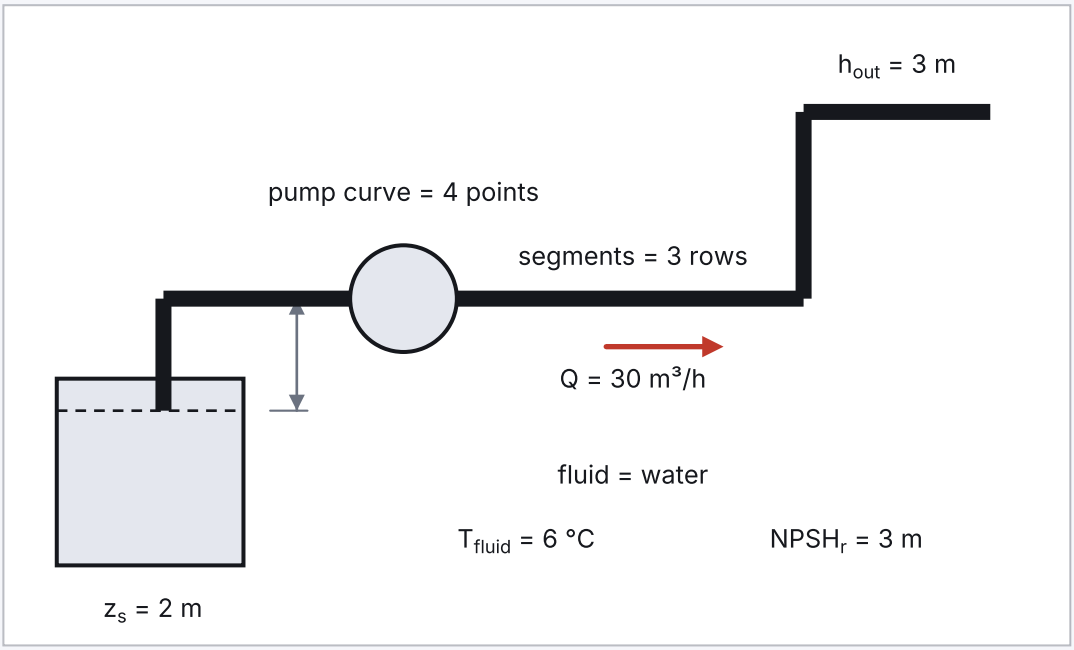


Figure 1 — A pump lifting from a suction level through the pipe segments as listed (three drawn) to an outlet that needs a pressure head. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
fluid	Fluid	water		spec: SPEC-M-04 §3: chilled water, no glycol	any
circuit	Circuit (a closed flow-and-return loop has no static lift: the rises entered are ignored)	open (the pump lifts the water by the rises entered)		default	any
segments	Pipe segments in series	3 row(s)		drawing: DRG-M-110 rev C	any
h_{out}	Pressure head required at the outlet	3	m	spec: Terminal unit: 30 kPa ≈ 3 m	0 … 1000 m
Q	Design flow rate (optional; blank = the flow comes from the heat load)	30	m³/h	spec: Schedule CHW-01 design flow	0.01 … 100000 m³/h
T_{fluid}	Fluid temperature (mean; link the mean water temperature of the sealed system for a heating loop)	6	°C	spec: SPEC-M-04 §3: 6 °C flow	0 … 100 °C
p_{atm}	Atmospheric pressure (NPSH)	101.325	kPa	assumption: Sea level.	50 … 110 kPa
z_s	Static head at the pump suction (+ flooded, − lift)	2	m	drawing: DRG-M-110 rev C	-10 … 100 m

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
v_{limit}	Velocity limit (warning and check)	2.5	m/s	spec: CIBSE Guide C practice limit, assumed	0.5 … 10 m/s
D_{law}	Law for the diameter ratio: a trimmed impeller of the same casing ($Q \propto D$, $H \propto D^2$) or a geometrically similar pump ($Q \propto D^3$, $H \propto D^2$)	impeller trim ($Q \propto n \cdot D$, $H \propto n^2 \cdot D^2$)		default	any
n_{ratio}	Speed ratio n_2/n_1 (affinity)	0.9		assumption: Inverter turn-down to 90 % speed.	0.2 … 2
D_{ratio}	Impeller diameter ratio D_2/D_1 (affinity; the law is chosen above)	1		assumption: Impeller unchanged.	0.7 … 1.1
η_{pump}	Pump efficiency at duty (for shaft power)	0.68		spec: Pump datasheet P-01 rev 2	0.2 … 0.95
Q_{pump}	Pump curve: flow points (optional; blank = no duty point)	[0, 20, 40, 60]	m³/h	spec: Pump datasheet P-01 rev 2, curve points	any
H_{pump}	Pump curve: head points (optional; blank = no duty point)	[40, 36, 24, 4]	m	spec: Pump datasheet P-01 rev 2, curve points	any
$NPSH_r$	NPSH required by the pump, at the flow $NPSH_a$ is evaluated at: the larger of the design flow and the duty flow (optional)	3	m	spec: Pump datasheet P-01 rev 2	0 … 50 m

TABLE ROW	VALUES
segments[1]	name = plantroom suction; L = 8 m; D = 100 mm; eps = 0.045 mm; K = 3; L_eq = 0 m; dz = 0 m; side = suction
segments[2]	name = riser DN80; L = 60 m; D = 80 mm; eps = 0.045 mm; K = 8; L_eq = 0 m; dz = 10 m; side = discharge
segments[3]	name = index run DN65; L = 45 m; D = 65 mm; eps = 0.045 mm; K = 12; L_eq = 6 m; dz = 2 m; side = discharge

3. Assumptions

A1. Flow is steady, incompressible and fully developed, and all segments carry the full design flow in series. (Darcy–Weisbach.)
A2. Fitting losses scale with velocity head (K method) or add pipe length (equivalent-length method). (Crane TP-410 practice.)
A3. Both fitting methods may be used on one segment. (Crane TP-410 practice.)
A4. The system curve scales turbulent and transitional friction and all fitting losses with Q^2 from the design flow. ($H_{sys} = H_{static} + k_1 Q + k_2 Q^2$.)
A5. The friction factor is held at its design-flow value on the system curve. ($H_{sys} = H_{static} + k_1 Q + k_2 Q^2$.)

A6. The friction of a laminar segment scales with Q, and the static part is independent of flow. ($H_{sys} = H_{static} + k_1 Q + k_2 Q^2$.)
A7. Water properties are from published correlations at atmospheric pressure, 0 to 100 °C. (Kell 1975; Vogel fit; Hyland & Wexler 1983.)
A8. Any other fluid is entered with its own density and viscosity and their provenance. (Stated.)
A9. p_{atm} (Atmospheric pressure (NPSH)) is taken as 101.325 kPa. (Sea level.)
A10. n_{ratio} (Speed ratio n_2/n_1 (affinity)) is taken as 0.9. (Inverter turn-down to 90 % speed.)
A11. D_{ratio} (Impeller diameter ratio D_2/D_1 (affinity; the law is chosen above)) is taken as 1. (Impeller unchanged.)

4. Working

[3] ρ = Kell (1975) density of water at T [4]
= T = 6 °C
= 999.94 kg/m³
Density
[4] μ = Vogel fit μ = 0.02939 exp(507.88/(T_K - 149.3)) mPa·s
= T_K = 279.15 K
= 1.4684 mPa·s
Dynamic viscosity
[5] p_v = Hyland–Wexler saturation pressure
= T = 6 °C
= 0.93525 kPa
Vapour pressure (NPSH)
[6] T_{line} = T (no T_{line} entered)
= 6 °C
= 6 degC
Design temperature of the line to insulate: the fluid temperature T, since no T_{line} was entered
[10] Q_{design} = Q
= 30 m³/h
= 30 m³/h
Design flow (as entered)
[11] v_{s1} = plantroom suction: Q / (π D²/4)
= 30 m³/h / (π × 100² mm² / 4)
= 1.061 m/s
Velocity, plantroom suction
[12] Re_{s1} = plantroom suction: ρ v D / μ
= 999.94 × 1.061 × 0.1 / 0.0014684
= 72253
Reynolds number, plantroom suction
[13] f_{s1} = plantroom suction: Darcy friction factor
= Colebrook–White, 10 iterations (Swamee–Jain start 0.021152)
= 0.021093
Friction factor, plantroom suction
[14] $f_{sj,s1}$ = plantroom suction: Swamee–Jain check
= 0.25 / log10(ε/3.7D + 5.74/Re^0.9)²
= 0.021152
Explicit cross-check, plantroom suction
[15] $h_{f,s1}$ = plantroom suction: f (L + L_{eq})/D · v²/2g + K v²/2g
= 0.021093 × (8 + 0) / 0.1 × 0.057399 + 3 × 0.057399
= 0.26906 m
Head loss, plantroom suction
[16] v_{s2} = riser DN80: Q / (π D²/4)
= 30 m³/h / (π × 80² mm² / 4)
= 1.6579 m/s

Velocity, riser DN80

[17] Re_{s2} = riser DN80: ρ v D / μ

= 999.94 × 1.6579 × 0.08 / 0.0014684

= 90316

Reynolds number, riser DN80

[18] f_{s2} = riser DN80: Darcy friction factor

= Colebrook-White, 10 iterations (Swamee-Jain start 0.020946)

= 0.020849

Friction factor, riser DN80

[19] $f_{sj,s2}$ = riser DN80: Swamee-Jain check

= 0.25 / log10(ε/3.7D + 5.74/Re^0.9)²

= 0.020946

Explicit cross-check, riser DN80

[20] $h_{f,s2}$ = riser DN80: f (L + L_{eq})/D · v²/2g + K v²/2g

= 0.020849 × (60 + 0) / 0.08 × 0.14014 + 8 × 0.14014

= 3.3123 m

Head loss, riser DN80

[21] v_{s3} = index run DN65: Q / (π D²/4)

= 30 m³/h / (π × 65² mm² / 4)

= 2.5113 m/s

Velocity, index run DN65

[22] Re_{s3} = index run DN65: ρ v D / μ

= 999.94 × 2.5113 × 0.065 / 0.0014684

= 111160

Reynolds number, index run DN65

[23] f_{s3} = index run DN65: Darcy friction factor

= Colebrook-White, 9 iterations (Swamee-Jain start 0.020964)

= 0.020834

Friction factor, index run DN65

[24] $f_{sj,s3}$ = index run DN65: Swamee-Jain check

= 0.25 / log10(ε/3.7D + 5.74/Re^0.9)²

= 0.020964

Explicit cross-check, index run DN65

[25] $h_{f,s3}$ = index run DN65: f (L + L_{eq})/D · v²/2g + K v²/2g

= 0.020834 × (45 + 6) / 0.065 × 0.32155 + 12 × 0.32155

= 9.1151 m

Head loss, index run DN65

[26] $h_{f,total}$ = Σ h_f

= 3 segment(s)

= 12.696 m

Total friction and fitting loss at the design flow

[28] V_{pipe} = Σ π/4·D²·L

= 3 segment(s)

= 513.75 L

Water held in the pipe segments

[29] H_{static} = Σ dz + h_{out}

= 12 + 3

= 15 m

Static head

[30] H_{sys} = H_{static} + $h_{f,total}$

= 15 m + 12.696 m

= 27.696 m

System head at the design flow

[31] dp_{total} = ρ * 9.80665 m/s^2 * $h_{f,total}$
= 999.94 kg/m³ × 9.80665 m/s^2 × 12.696 m
= 124.5 kPa
Friction pressure drop at the design flow
[32] dp_{sys} = ρ * 9.80665 m/s^2 * H_{sys}
= 999.94 kg/m³ × 9.80665 m/s^2 × 27.696 m
= 271.59 kPa
System pressure at the design flow
[33] P_{hyd} = ρ * 9.80665 m/s^2 * Q_{design} * H_{sys}
= 999.94 kg/m³ × 9.80665 m/s^2 × 30 m³/h × 27.696 m
= 2.2633 kW
Hydraulic power at the design point
[40] Q_{duty} = duty flow (as entered)
= $H_{pump}(Q)$ = $H_{sys}(Q)$, scan and bisection
= 31.569 m³/h
Duty flow as entered
[41] H_{duty} = duty head (as entered)
= H_{sys}(31.569 m³/h)
= 29.059 m
Duty head as entered
[42] P_{shaft} = ρ g Q H / η
= 999.94 × 9.8066 × 0.008769 × 29.059 / 0.68
= 3.6747 kW
Shaft power at duty as entered
[43] affinity = $Q_2 = Q_1 (n_2/n_1)(D_2/D_1)$, $H_2 = H_1 (n_2/n_1)^2(D_2/D_1)^2$ (impeller trim)
= n_2/n_1 = 0.9, D_2/D_1 = 1
= "curve scaled"
Affinity laws applied to the pump curve
[44] Q_{duty2} = duty flow (after the affinity shift)
= $H_{pump}(Q)$ = $H_{sys}(Q)$, scan and bisection
= 26.238 m³/h
Duty flow after the affinity shift
[45] H_{duty2} = duty head (after the affinity shift)
= H_{sys}(26.238 m³/h)
= 24.712 m
Duty head after the affinity shift
[46] P_{shaft2} = ρ g Q H / η
= 999.94 × 9.8066 × 0.0072883 × 24.712 / 0.68
= 2.5972 kW
Shaft power at duty after the affinity shift
[49] $h_{f,suction}$ = Σ h_f on the suction side
= 1 suction segment(s)
= 0.26906 m
Suction-side friction loss
[50] $NPSH_a$ = p_{atm} / (ρ * 9.80665 m/s^2) + z_s - $h_{f,suction}$ - p_v / (ρ * 9.80665 m/s^2) (Ch. 8)
= 101.32 kPa / (999.94 kg/m³ × 9.80665 m/s^2) + 2 m - 0.26906 m - 0.93525 kPa / (999.94 kg/m³ × 9.80665 m/s^2)
= 11.968 m
NPSH available
[52] u_{npsh} = $NPSH_r$ / $NPSH_a$
= 3 m / 11.968 m
= 0.25066
NPSH utilisation (required / available)

[53] v_{peak} = highest segment velocity
= 2.5113 m/s
= 2.5113 m/s
Highest velocity
[54] u_v = v_{peak} / v_{limit}
= 2.5113 / 2.5
= 1.0045
Velocity utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
Q_{design}	30 m³/h	Design flow
H_{static}	15 m	Static head
V_{pipe}	513.7 L	Water held in the pipes
H_{sys}	27.70 m	System head at Q
$h_{f,total}$	12.70 m	Σ h_f
dp_{total}	124.5 kPa	Friction Δp
dp_{sys}	271.6 kPa	System Δp
P_{hyd}	2.263 kW	Hydraulic power
v_{peak}	2.512 m/s	Highest velocity (governing)
T_{line}	6 °C	Line design temperature
$NPSH_a$	11.97 m	NPSH available
Q_{duty}	31.57 m³/h	Duty flow as entered
H_{duty}	29.06 m	Duty head as entered
P_{shaft}	3.675 kW	Shaft power as entered
v_{s1}	1.061 m/s	Velocity, plantroom suction
Re_{s1}	72253	Re, plantroom suction
f_{s1}	0.021093	f, plantroom suction
$f_{sj,s1}$	0.021152	f (Swamee–Jain), plantroom suction
$h_{f,s1}$	0.2691 m	h_f, plantroom suction
v_{s2}	1.658 m/s	Velocity, riser DN80
Re_{s2}	90316	Re, riser DN80
f_{s2}	0.020849	f, riser DN80
$f_{sj,s2}$	0.020946	f (Swamee–Jain), riser DN80
$h_{f,s2}$	3.312 m	h_f, riser DN80
v_{s3}	2.511 m/s	Velocity, index run DN65
Re_{s3}	111160	Re, index run DN65
f_{s3}	0.020834	f, index run DN65
$f_{sj,s3}$	0.020964	f (Swamee–Jain), index run DN65

SYMBOL	VALUE	DESCRIPTION
$h_{f,s3}$	9.115 m	h_f, index run DN65
Q_{duty2}	26.24 m³/h	Duty flow after the affinity shift
H_{duty2}	24.71 m	Duty head after the affinity shift
P_{shaft2}	2.597 kW	Shaft power after the affinity shift

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Highest velocity against limit	2.512 m/s	2.500 m/s	1.01	
PASS	NPSH required against NPSH available	3 m	11.97 m	0.25	
PASS	Duty point against the system curve (as entered)	31.569 m³/h	29.059 m	–	
PASS	Duty point against the system curve (after the affinity shift)	26.238 m³/h	24.712 m	–	

7. Applicability

WARNING segments: index run DN65 runs at 2.5113 m/s, above the 2.5 m/s limit. [velocity]

Defaults used (inputs left empty):
circuit: not entered, so the default of 'open (the pump lifts the water by the rises entered)' is used.
D_{law}: not entered, so the default of 'impeller trim (Q ∝ n·D, H ∝ n²·D²)' is used.

8. Verification

```
Template bs.pipe-pump@1.10.1 Content hash 2df24301b0231856 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. Crane Co. Flow of Fluids Through Valves, Fittings and Pipe. Technical Paper No. 410. Crane Co.
2. Çengel YA, Cimbala JM. Fluid Mechanics: Fundamentals and Applications. 4th ed. McGraw-Hill Education; 2018.
3. Chartered Institution of Building Services Engineers. CIBSE Guide C: Reference Data. CIBSE; 2007.
4. Kell GS. Density, thermal expansivity, and compressibility of liquid water from 0° to 150°C: correlations and tables for atmospheric pressure and saturation reviewed and expressed on 1968 temperature scale. J Chem Eng Data. 1975;20(1):97-105.